

International Islamic University Chittagong
Center for General Education (CGED)
 Midterm Examination, Autumn-2025

Course Code: GEED-2302 Course Title: Sciences of Qur'an and Hadith
(For Law faculty: GEED-2101)

Full Marks: 30

Time: 1 hour & 30 minutes

Answer all questions. The right side columns contain marks, CLOs, and Bloom's taxonomy domain for each question.

#	Questions	Marks	CLOs	Bloom's taxonomy domain
1	(a) Explain briefly (any four) • Definition of Sciences of <i>Qur'an</i> • Five main names of the holy <i>Qur'an</i> • Subject matter of the holy <i>Qur'an</i> • First Revelation of the holy <i>Qur'an</i> • Numbers of Ayah of the holy <i>Qur'an</i>. Or, (b) Define Al- <i>Qur'an</i> literally and terminologically. Explain the necessity of the holy <i>Qur'an</i> with proper evidence.	10	2	Create
2	"The holy <i>Qur'an</i> was revealed in stages"- justify the statement explaining the wisdom behind gradual revelation.	10	2	Evaluate & Create
3	"The arrangement of <i>Ayah</i> and <i>Surah</i> of the holy <i>Qur'an</i> is by the command of Prophet (SAAS)"- Prove the statement defining <i>Ayah</i> and <i>Surah</i> literally and terminologically.	10	2	Remember & Evaluate

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International Islamic University Chittagong

Department of Computer Science & Engineering

Program: B.Sc. in CSE: Semester: 3^r

Mid Term Special Examination, Autumn-2025

Course Code: CSE-2323

Course Title: Digital Logic Design

Time: 1 Hour 30 minutes.

Total Marks: 30

Answer the following Three (3) questions. Each question carries 10 marks.

Question No	Question	CLOs	Marks
1.a	Design basic gates (AND, OR, NOT) using only NAND and NOR gates. Show the necessary truth tables, circuit diagrams, and Boolean equations.	CLO1	5
1.b	Design a Full Adder circuit using two Half-Adders and An OR gate and Explain how the Sum and Carry-Out functions of a Full Adder are derived using the XOR operation.	CLO2	5
2.a	Simplify the Boolean function using K-Map: $F(w, x, y, z) = \sum(1,3,7,11,15)$ and $d(w, x, y, z) = \sum(0,2,5)$	CLO2	4
2.b	Express the Boolean functions: I. $F = A + B'C$ in Sum of Minterm. (SOP) II. $F = xy + x'z$ in a Product of Maxterm.(POS)	CLO1	4
2.c	Using Hamming code (even parity) , determine the position of the error in the received 7-bit code word 1011010 .	CLO1	2
3.a	Simplify the following Boolean expressions to a minimum number of literals. I. $x'y' + xy + x'y$ II. $ABC + A'B + ABC'$	CLO2	4
OR			
3.c	Design a 4-bit combinational circuit with inputs A, B, C, D and output F, where $F = 1$ if the number is divisible by 3, otherwise 0. Derive the truth table, simplify the Boolean expression using a K-map, and implement the logic using AND, OR, and NOT gates.	CLO2	4
3.b	For the Boolean functions E and F , as given in the following truth table: I. List the minterms(SoP) and maxterms(PoS) of each function. II. List the minterms(SoP) of $E+F$ and $E \cdot F$. III. Simplify E and F to expressions with a minimum of literals	CLO2	6

X	Y	Z	E	F
0	0	0	0	1
0	0	1	1	0
0	1	0	1	1
0	1	1	0	0
1	0	0	1	1
1	0	1	0	0
1	1	0	1	0
1	1	1	0	1

International Islamic University Chittagong
Department of Computer Science and Engineering
B.Sc. in CSE Midterm Examination, Autumn 2025
Course Code: STAT 2311 Course Title: Probability and Statistics
Total marks: 30, Time: 1 hour 30 minutes

[Answer all the questions; Figures in the right-hand margin indicate full marks.]

CO DL

1(a) Define Statistics as a subject and explain its connection with Computer Science and Engineering. 4 CO1 C3

1(b) Define a variable. Mention its types. Classify the scale of measurement of each of the following terms: Download speed, Network protocol used, Performance rating, IQ scores, distance traveled, and Proficiency in coding. 6 CO1 C4

2(a) Explain the objectives of using measures of central tendency in data preprocessing. Discuss with an example when the geometric mean cannot be calculated. 4 CO1 C3

2(b) A database administrator is monitoring the response times of queries in a database system. The response times (in milliseconds) are shown into the following frequency table: 6 CO1 C6

Response Time (ms)	5–10	10–15	15–20	20–25	25–30
Frequency	3	7	13	5	2

(i) Calculate the median, third quartile, and 70th percentile from the given data.

(ii) Draw an appropriate graph to locate 1st quartile and 85th percentile.

3(a) What are the various types of measures of dispersion in statistics? “Relative measures of dispersion are more suitable for comparing two or more data sets” – why? 4 CO1 C2

3(b) The exam scores of five students in class A and class B are given below: 6 CO1 C3

Student	1	2	3	4	5
Class A	50	55	60	65	65
Class B	20	40	50	70	80

(i) Which class has the higher average exam score?

(ii) Which class of students has more consistent exam scores?

Or

a) Explain the following terms with example (any two): 4 CO1 C2
 (i) Population, (ii) Primary data, (iii) Questionnaire

b) The following data represents the number of children in seven families: 2, 0, 5, 3, 4, 6, 1. Calculate the arithmetic mean and standard deviation for the above dataset. What happens to your measures if a fixed number, such as 2, is added to the number of children in each family? 6 CO1 C3

International Islamic University Chittagong (IIUC)
 Department of Computer Science and Engineering (CSE)
 B. Sc. in CSE, Mid Term Examination, Autumn-2025
 Course Code: MATH-2307, Course Title: Mathematics-III

Time: 1:30 hours

Marks: 30

[Answer all questions; Read the instruction below before answering the questions; Figures in right hand margin indicate full marks]

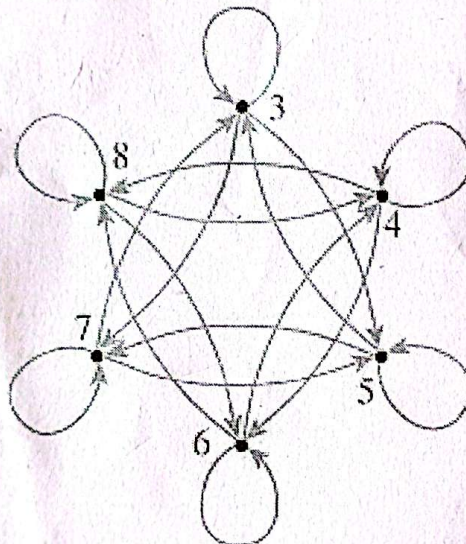
1. a) If the matrix $A = \begin{bmatrix} 4 & x+6 \\ 2x+3 & x+1 \end{bmatrix}$ is symmetric, find the value of x 2 CLO1 C2

Or

- Check whether it is singular matrix $\begin{bmatrix} 1 & 2 & d \\ 2 & 4 & e \\ 3 & 5 & a \end{bmatrix}$; Where a, d and f are the digits of your own ID number 2 CLO1 C2

- b) Find Inverse of the matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ by making Linear Equations 3 CLO1 C2

- c) 3 CLO1 C2



Create an adjacency matrix for the above relation

- d) When do we use free variables when solving linear systems of equations? Give an example. 2 CLO1 C2

Or

- Test whether $\lambda_1 = f$ and $\lambda_2 = e$ are Eigen values for $A = \begin{bmatrix} 5 & 3 \\ 2 & 3 \end{bmatrix}$; Where e and f are the digits of your own ID number 2 CLO2 C2

2. a) Determine which of the following vectors are Eigen vectors for 5 CLO2 C2

$$A = \begin{pmatrix} 2 & e \\ -1 & -5 \end{pmatrix} \text{ showing your analysis procedure graphically}$$

(a) $\begin{pmatrix} a \\ f \end{pmatrix}$ (b) $\begin{pmatrix} b \\ c \end{pmatrix}$; Where a, b, e and f are the digits of your own ID number

b) Determine whether the following vectors in R^3 are linearly dependent 5 CLO2 C2

$$\text{or linearly independent } x_1 = \begin{bmatrix} 2 \\ 2 \\ 0 \end{bmatrix}, x_2 = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}, x_3 = \begin{bmatrix} 4 \\ 2 \\ -2 \end{bmatrix}$$

Or

Verify Cayley Hamilton theorem for the matrix $\begin{bmatrix} 4 & 1 \\ 1 & 4 \end{bmatrix}$ and hence find A^{-1} . 5 CLO2 C2

3. Diagonalize the matrix $A = \begin{bmatrix} 1 & 6 \\ 2 & 5 \end{bmatrix}$ 10 CLO2 C2

Instruction:

If your ID no is C171023 then you can assign your digit as follows

C	1	7	1	0	2	3
	↓	↓	↓	↓	↓	↓
	a	b	c	d	e	f

To solve the above problems, you need to use your own ID number where needed

International Islamic University Chittagong (IIUC)

Department of Computer Science and Engineering

Midterm Examination

Program: **B.sc (Eng.)**
 Course Code: **CHEM-2301**
 Total Marks: **30**

Semester: **Autumn- 2025**
 Course Title: **Chemistry**
 Time: **1 Hour 30 Minutes**

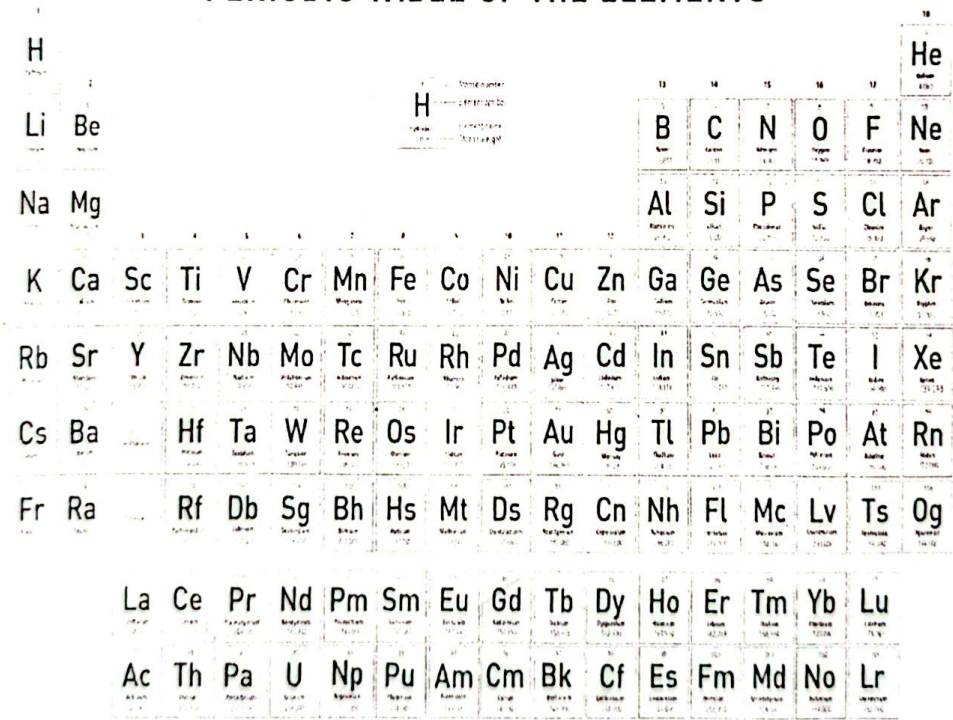
- (i) Answer all the questions. The figures in the right-hand margin indicate full marks.
 (ii) Course Learning Outcomes (CLOs) and Bloom's Levels are mentioned in additional Columns.

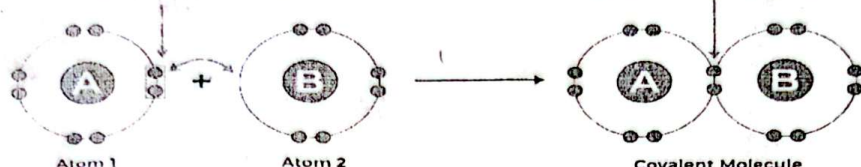
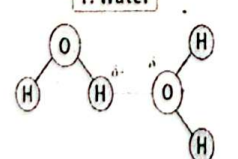
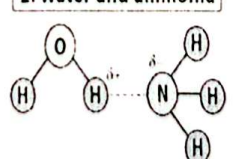
Course Learning Outcomes (CLOs) of the Questions

- CLO1** Developing the basic knowledge of different chemicals.
- CLO2** Understanding of the chemical behaviour and physical properties of common substances.
- CLO3** Demonstrate the basic proficiency to solve the chemical problems and discuss the interactions between matter and energy at the atomic and molecular levels.

Bloom's Levels of the Questions

Letter Symbols	R	U	Ap	An	E	C
Meaning	Remember	Understand	Apply	Analyze	Evaluate	Create

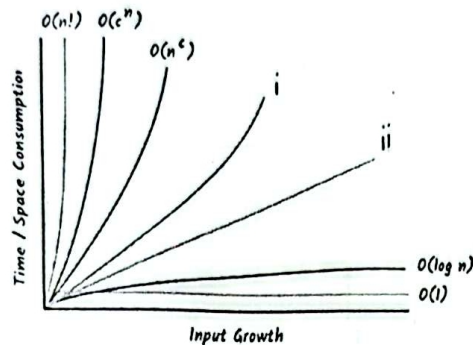
Q1.	3)	PERIODIC TABLE OF THE ELEMENTS 																		
		State the Periodic Law showed in figure. Mention it's advantages. "The orbitals are filled up with electrons in the increasing order of their energy"- Justify this statement with above periodic table.																CLO1	An	5
	4)	Point out periodic properties. How atomic size, metallic characteristics, oxide nature and electronegativity varying within Periods and groups? Describe with example.																CLO1	R	5

Q2.	a)	Share your idea about chemical bond. Evaluate different types of chemical bonds with suitable examples.	CLO2	E	5												
	b)	Pair of Valence Electrons Available for Bonding Sharing of Available Electrons Coming from A  Atom 1 Atom 2 Covalent Molecule Which bond mentioned in the above picture? Explain about it. State hybridization. Show the hybridization of Methane (CH₄).	CLO2	Ap	5												
Q3.	a)	Distinguish between Mendeleev's periodic table and Modern periodic table. What are the disadvantages of Mendeleev's periodic table?	CLO1	U	5												
	b)	Define groups and blocks in the modern periodic table. Briefly explain "p block" elements. Ionization Energies of Al <table><thead><tr><th>Ionization</th><th>IE (kJ/mol)</th></tr></thead><tbody><tr><td>$\text{Al} \longrightarrow \text{Al}^+ + e^-$</td><td>$IE_1 = 578$</td></tr><tr><td>$\text{Al}^+ \longrightarrow \text{Al}^{2+} + e^-$</td><td>$IE_2 = 1,817$</td></tr><tr><td>$\text{Al}^{2+} \longrightarrow \text{Al}^{3+} + e^-$</td><td>$IE_3 = 2,745$</td></tr><tr><td>$\text{Al}^{3+} \longrightarrow \text{Al}^{4+} + e^-$</td><td>$IE_4 = 11,575$</td></tr><tr><td>$\text{Al}^{4+} \longrightarrow \text{Al}^{5+} + e^-$</td><td>$IE_5 = 14,830$</td></tr></tbody></table> "Changes in ionization energy depend on various factors."-Justify.	Ionization	IE (kJ/mol)	$\text{Al} \longrightarrow \text{Al}^+ + e^-$	$IE_1 = 578$	$\text{Al}^+ \longrightarrow \text{Al}^{2+} + e^-$	$IE_2 = 1,817$	$\text{Al}^{2+} \longrightarrow \text{Al}^{3+} + e^-$	$IE_3 = 2,745$	$\text{Al}^{3+} \longrightarrow \text{Al}^{4+} + e^-$	$IE_4 = 11,575$	$\text{Al}^{4+} \longrightarrow \text{Al}^{5+} + e^-$	$IE_5 = 14,830$	CLO2	An	5
Ionization	IE (kJ/mol)																
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OR																	
Q3.	a)	List out the name of some transition metals. Compare between transition metals, lanthanides and actinides. How ion dipole interaction occurred?	CLO2	Ap	5												
	b)	1. Water  2. Water and ammonia  What kind of bond available in the figure? Discuss about it's importance.	CLO2	An	5												

Bismillahir Rahmanir Rahim
International Islamic University Chittagong
 Department of Computer Science & Engineering
Mid Term Examination, Autumn 2025
CSE 2321 Data Structures
 Total marks: 30 Time: 90 minutes

[Answer *all* of the following questions. Figures in the right-hand margin indicate full marks.]

1. a) Explain why Big O notation is preferred over measuring algorithm time in seconds. 1
 Find the Time complexity of (i) and (ii) on the following graph. +
2



- b) Draw a flowchart or write a code for linear search algorithm. 2
[OR]
 Draw a flowchart or write a code for finding the second largest element of an array.

- c) i) Given the pattern $P = \text{'aac'}$ and the text $T = \text{'abaaaccacaaaxx'}$, determine the number N of character comparisons required to find the **first occurrence** (index) of P in T using the **naïve pattern matching algorithm**. 3
+
 Show the matching process step by step **until the pattern is found**. 1
+

ii) State the **time complexity** of the pattern-matching algorithm you used. 1

iii) Give two pairs of examples of pattern P and text T , where, pattern length $|P| = 4$ and text length $|T| = 10$:

- One pair gives the **minimum** number of character comparisons.
- One pair gives the **maximum** number of character comparisons.

2. a) Explain how to represent a two-dimensional array in computer memory using row-major order and column-major order. Illustrate your answer with an appropriate example or diagram. 3

- b) Consider a 2D array A (4:12, 7:14). 3
 i. Find the length of each dimension and the total number of elements in A .
 ii. If $\text{Base}(A) = 300$ and $w = 8$ bytes per memory cell, calculate the address of the element $A[6, 10]$ assuming row-major storage order.

- c) Given a string **CANDLE**, find the number of *comparisons* (C) and number of *interchanges* (D) needed to *sort* the string *alphabetically* by using *bubble sort* algorithm. Show each steps. 4

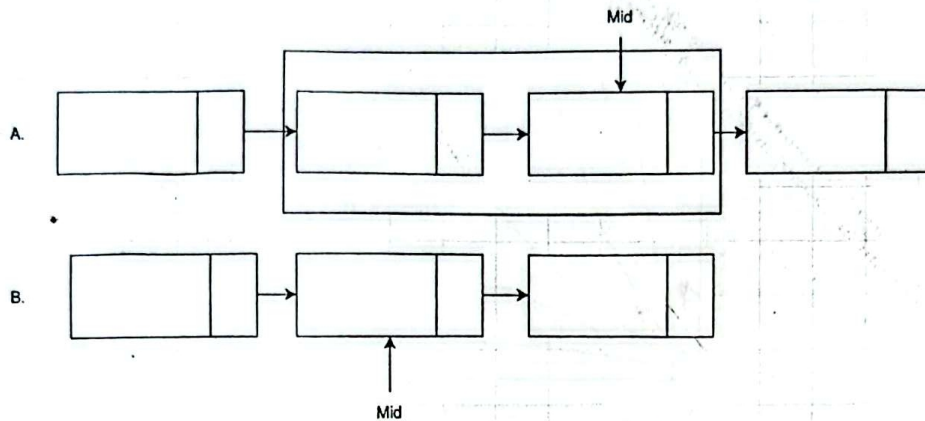
[OR]

What is the time complexity of the binary search algorithm?

Show the intermediate steps to search the $\text{ITEM} = 41$ and $\text{ITEM} = 97$ from the following list of integers using binary search algorithm.

21, 25, 30, 35, 41, 45, 53, 66, 68, 71, 82, 89, 101

3. a) What is the difference between Array and Linked List? Which one is better for searching an element and why? 2
- b) (i) Write an algorithm or a program to insert a node in the middle position of a Linked List. Your algorithm/program should handle the following cases: 3
1. When node number is even, return as stated in A. +
 2. When node number is odd, return as stated in B. 1



(ii) Explain how the same insertion can be achieved by traversing the linked list only once.

[OR]

You are managing a reading list using a **doubly linked list**. Each node stores a book title. The list has 5 books with the following addresses:

Book Title	Node Address	Previous Address	Next Address
Book C	1502	3071	4096
Book A	2005	NULL	3071
Book E	2810	4096	NULL
Book B	3071	2005	1502
Book D	4096	1502	2810

Tasks:

1. Draw the **doubly linked list** showing nodes, data, previous and next pointers.
2. Insert a new book "**Book F**" after "**Book C**" and show the updated links.
3. Remove "**Book B**" from the reading list and update the links accordingly

- 4) Write an algorithm or program to insert a node into a singly linked list at a given position (*pos*, starting from 1). Your solution should:

- Insert in **O(1)** time if inserting at the beginning or end
- Insert in **O(n)** time when inserting at any intermediate position.
- Handle invalid positions appropriately.